

ViLLa: Video Reasoning Segmentation with Large Language Model

Rongkun Zheng¹ Lu Qi² Xi Chen¹ Yi Wang^{3,4}
 Kun Wang⁵ Hengshuang Zhao^{1*}

¹The University of Hong Kong ²Insta360 ³Shanghai Artificial Intelligence Laboratory

⁴Shanghai Innovation Institute ⁵SenseTime Research
 {zrk22@connect, hszhao@cs}.hku.hk

Abstract

*Recent efforts in video reasoning segmentation (VRS) integrate large language models (LLMs) with perception models to localize and track objects via textual instructions, achieving barely satisfactory results in simple scenarios. However, they struggled to discriminate and deduce the objects from user queries in more real-world scenes featured by long durations, multiple objects, rapid motion, and heavy occlusions. In this work, we analyze the underlying causes of these limitations, and present **ViLLa: Video reasoning segmentation with Large Language Model**. Remarkably, our ViLLa manages to tackle these challenges through multiple core innovations: (1) a context synthesizer that dynamically encodes the user intent with video contexts for accurate reasoning, resolving ambiguities in complex queries, and (2) a hierarchical temporal synchronizer that disentangles multi-object interactions across complex temporal scenarios by modelling multi-object interactions at local and global temporal scales. To enable efficient processing of long videos, ViLLa incorporates (3) a key segment sampler that adaptively partitions long videos into shorter but semantically dense segments for less redundancy. What's more, to promote research in this unexplored area, we construct a VRS benchmark, *VideoReasonSeg*, featuring different complex scenarios. Our model also exhibits impressive state-of-the-art results on *VideoReasonSeg*, *Ref-YouTube-VOS*, *Ref-DAVIS17*, *MeViS*, and *ReVOS*. Both quantitative and qualitative experiments demonstrate that our method effectively enhances video reasoning segmentation capabilities for multimodal LLMs.*

1. Introduction

Capitalizing on the achievements of Large Language Models (LLMs) [10, 38, 39, 75], the development of large multimodal models (LMMs) [3, 25, 32, 34, 58, 80] has notably enhanced visual perception capabilities and user-interaction

experiences to new heights. However, in the absence of explicit instructions specifying target objects or categories, the majority of models [1, 24, 25, 31, 32, 71, 80], including both perception models and language ones, struggle to generate detailed and granular outputs such as instance masks. Instead, these models typically generate only general textual descriptions for images or videos. This limitation hinders the practical application of multimodal systems in various scenarios, such as autonomous driving [57], image and video editing [30], robotics [48], and augmented reality [46].

Recent studies [23, 45, 70] have investigated the application of LLMs to generate object masks in a novel reasoning segmentation task that requires complex reasoning to interpret sophisticated instructions. These approaches introduce efficient language models capable of producing multiple open-set targets while accommodating diverse reasoning complexities. However, despite their successes, these methods remain limited to image-level reasoning and do not extend fine-grained reasoning capabilities to the temporal dimension. To address this issue, researchers [4, 67] managed to equip the image reasoning segmentation model with temporal understanding ability, and propose VRS. VISA [67] employs a video understanding model [28] to select the key frame for LISA [23] to segment and subsequently utilizes an object tracker [8] to maintain temporal consistency. However, the quality of segmentation is highly dependent on the accuracy of key frame selection, which is susceptible to error accumulation through propagation as the number of input frames increases. On the other hand, VideoLISA [4] adopts a sparse dense sampling strategy to compress visual features and extends the single segmentation token to track an instance across frames, especially when the movement is rapid. Nevertheless, the sampling strategy fails to adaptively select the key video segments, making it challenging to capture the essential information from the video. Also, because a single segmentation token is unable to segment and track multiple instances across frames, the segmentation and tracking quality decrease significantly when the video scene becomes intricate (e.g., involving multiple objects with occlusion),

*Corresponding author.

as proven in various previous Video Instance Segmentation (VIS) approaches [20, 78], which utilize multiple-level of tokens to segment and track instances.

To tackle the aforementioned challenges, we propose **ViLLa**, an effective and efficient LMM for video reasoning segmentation. ViLLa proficiently perceives and reasons videos with multiple objects, rapid motion, and diverse reasoning complexities (as shown in Fig. 1). Specifically, our ViLLa has three key components: Key Segment Extractor, Context Synthesizer, and Hierarchical Temporal Synchronizer. These designs put effort into solving the complex video reasoning segmentation from different perspectives. The Key Segment Extractor selects the most relevant video segments, which reduces redundant visual tokens and effectively turns long videos into shorter segments. To make the responses more pertinent to users' intentions, Context Synthesizer synthesizes frame context information from visual features to text embeddings to generate contextually enriched text and highly input-responsive visual embeddings. To build up stronger temporal relations to multi-target segmentation tokens in complex scenes, the Hierarchical Temporal Synchronizer communicates and consolidates the multi-scale segmentation embeddings across temporal dimension via attention-based fusion strategy. Tested on multiple referring video object segmentation benchmarks, our model could surpass previous state-of-the-art LLM-based methods by a large margin. In addition, to validate its effectiveness, we establish a benchmark for video reasoning segmentation evaluation, called VideoReasonSeg. Utilizing a GPT-4V [39]-aided data curation pipeline, we curate 3k high quality video samples for video reasoning segmentation spanning multiple-choice QAs and video-instruction-mask samples, and the video scenarios contain diverse reasoning complexities (as presented in the appendix), which offers convincing metrics.

In summary, our contributions are as follows:

- We analyze the limitations of current VRS approaches and propose a framework named ViLLa that segments and tracks the desired objects in more complex scenarios.
- We develop three key modules that resolves complex scenario reasoning from different perspectives, namely key segment extractor, context synthesizer, and hierarchical temporal synchronizer. Our extractor selects the most relevant video segments from the complex input video, while the synthesizer combines and infuses user intention and context information, and the synchronizer communicates multi-scale segmentation tokens spatio-temporally.
- We develop VideoReasonSeg, a comprehensive benchmark for VRS that spans different complex reasoning scenarios, comprising 3k videos and 15k object descriptions for instruction tuning and evaluation. This benchmark plays a crucial role in evaluating and fostering the exploration of reasoning capabilities in video-based models.
- We conduct extensive experiments on challenging refer-

ring segmentation benchmarks, ReVOS, and our proposed VideoReasonSeg. The state-of-the-art results demonstrate the effectiveness of our method.

2. Related Works

Large Multimodal Models. Large multimodal models (LMMs) can be broadly categorized into two groups based on their utilization of large language models (LLMs). The first category comprises models [35, 56, 74] that are either trained from scratch or leverage smaller language models like BERT for text processing. These models typically employ a combination of contrastive and generative objectives to tackle a range of multimodal tasks (e.g., Coca [74]). However, their limited language understanding capacity often restricts their performance in tasks that demand massive commonsense reasoning abilities.

The emergence of LLMs in recent years has paved the way for a new paradigm in the development of LMMs, in which LLMs are enhanced with multimodal comprehension abilities [40, 50, 51, 76]. This approach typically includes the use of adapters to align visual and textual representations within LLMs, as exemplified by Flamingo [1], BLIP-2 [25], MiniGPT-4 [80], LLaMA-Adapter [16], LLaVA [32], InstructBLIP [11], InternGPT [34], QwenVL [3], InternVideo2 [59]. Some video-related MLLM (VideoLLM) VideoChatGPT [37], and Valley [36] utilize ChatGPT [38] to generate video instruction-tuning data, aiming to enhance instruction-following capabilities for real-world video comprehension. Even though these models have demonstrated improved performance in vision-language tasks, their primary limitation lies in generating only textual outputs.

Video Segmentation. Video Instance Segmentation (VIS) [69] aims to simultaneously detect, segment, and track object instances in videos based on predefined object categories. Numerous studies [26, 29, 60, 63, 79] have proposed diverse designs (such as tracking by detection [2], masked attention [7], object token association [20, 27, 65], contrastive memory [19, 72], and more) to effectively model temporal relationships. For example, IDOL [66], which is built upon Deformable-DETR [81], incorporates a contrastive learning head for associations [17]. DVIS [78] proposed a decoupling strategy by dividing VIS into image segmentation, tracking, and refinement. The Referring Video Object Segmentation (RVOS) task was introduced to segment the target instance referred to in a given text. Early works [21, 47] used the spatial-temporal memory mechanism to strengthen temporal relations. The researchers then started building offline reference models [53, 54, 61, 68] that take the whole clip as input. Currently, query-based Transformer models [5, 64] begin to dominate. MTTR [5] utilized an instance-level segmentation pipeline to determine the most suitable sequence for the referred object. ReferFormer [64] transformed text expressions into queries to attend to relevant regions.

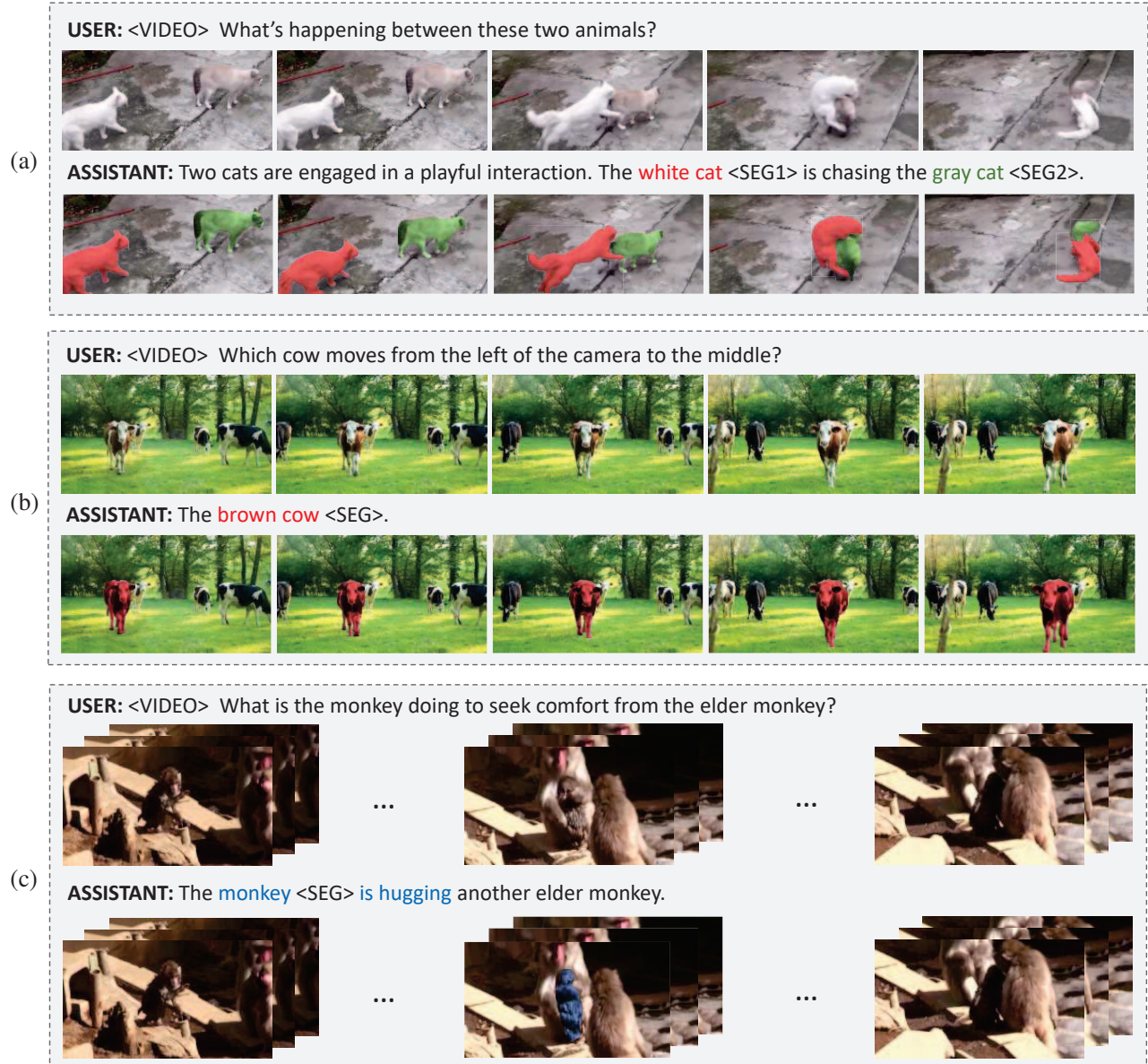


Figure 1. Our ViLLa is an effective and efficient LMM capable of segmenting and tracking: (a) multiple objects with rapid motion; (b) objects in crowded scenes; (c) objects in long videos with occlusions.

Reasoning Segmentation. In numerous practical applications, it is essential to comprehend visual inputs at a fine-grained level, such as specific regions. Kosmos-2 [41], InternGPT [34], and Ferret [73] offer grounding capabilities to specified image regions. Yet, these approaches still fail to provide pixel-level outputs. To bridge this gap, LISA [23] integrates SAM [22] with LLMs to tackle segmentation tasks. PixelLM [45] proposes a novel pixel decoder and a segmentation codebook to generate masks for multiple targets. Then, GLaMM [44] proposes region-level caption and segmentation tasks. Several works [14, 33, 43] use LLMs as agents to assign different visual experts. OMG-LLaVA [77] imple-

ments a new comprehensive baseline that has only a single visual encoder. However, these methods mainly focus on reasoning about images and fail to reason across temporal dimensions. When they are transferred to the video domain, the segmentation token is unable to model multiple targets as well as their trajectories. Early approaches (VISA [67] and VideoLISA [15]) manage to bridge the gap by implementing different frame sampling strategies, but VISA suffers from the error accumulation from key frame selection, and VideoLISA fails to adaptively extract relevant frames which leads to redundancy. After analyzing their drawbacks, we propose ViLLa to address the problem.

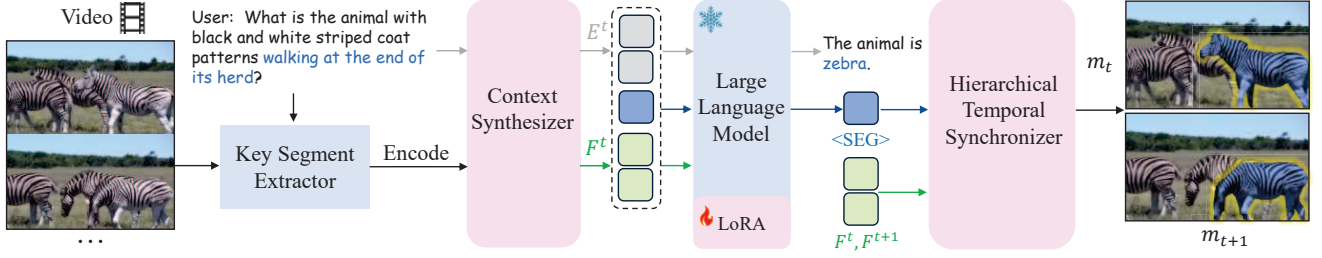


Figure 2. Overall framework of ViLLa. Given the input video frames and user input text query, the Key Segment Extractor is proposed to select the most query-relevant video segments from the video. Then, the Context Synthesizer aggregates text-related visual cues from the visual features to the text embeddings and selects the relevant visual features. The large language model generates text output and segmentation tokens with the input of visual features, visually enriched text embeddings, and pre-defined segmentation tokens. Finally, the segmentation tokens are fed to the Hierarchical Temporal Synchronizer to produce the final output segmentation mask tracklets.

3. ViLLa

Video reasoning segmentation aims to output a binary segmentation mask sequence $\{m_t\}_{t=1}^T \in \mathbb{R}^{T \times H \times W}$, given an input video clip consisting of T frames $V \in \mathbb{R}^{T \times H \times W \times 3}$ and a query instruction $\mathbf{x}_{txt}$. The task shares a similar formulation with the referring video object segmentation (VOS), but it has complex queries that require reasoning.

3.1. Model Design

Model Overview. As depicted in Fig. 2, ViLLa features a streamlined architecture, comprising mainly several main parts: 1) a pre-trained visual encoder that aligns with a text encoder, 2) a large language model (LLM), 3) a key segment extractor, 4) a context synthesizer, and 5) a hierarchical temporal synchronizer.

The key segment extractor aims to extract the pivotal video segments from a long video input. The context synthesizer and the hierarchical temporal synchronizer are crucial in equipping the LLMs with the capacity to generate masks across temporal dimensions. We utilize the context synthesizer to aggregate the target-text-relevant visual features of the current frame to the input text embeddings. The output text embeddings that carry the most crucial visual cues are then fed into the LLM along with the visual features. In the hierarchical temporal synchronizer, we use the video-scale segmentation tokens to interact with frame-level tokens to produce the final refined complete segmentation tokens that contain both video-level and frame-level information. Finally, the segmentation tokens produce the prediction masks in conjunction with the multi-scale video features.

Encoder. For input frame V_t , the visual encoder $\mathcal{I}$ extracts multi-scale visual features $F_t = \{F_t^\ell\}_{\ell=1}^L$ from $\mathcal{I}(V_t)$, comprising L visual features at the selected layers of $\mathcal{I}$. Meanwhile, the user instruction is fed into the text encoder with the visual embedding $F_t^L \in \mathbb{R}^{N \times C}$ (where $N = H/p \times W/p$ and C indicate the number of frame patches and embedding channels, respectively) and gener-

ates the $\mathbf{x}_{txt} \in \mathbb{R}^{M \times C}$, where M denotes the number of queries. We choose QFormer as the text encoder to maintain the cross-modality alignment. By incorporating this approach, the text embeddings now capture visual cues that are highly relevant to the user’s instructions. The output of the final layer, F_t^L , encodes global image information and is transformed to align with the language space of LLMs via a vision-to-language projection layer $\text{Proj}_{V \rightarrow L}$.

3.2. Key Segment Extractor

For short videos that typically has very few actions and objects, the segmentation can be solved by image-level reasoning segmentation methods in a frame-by-frame style. These cases, however, can’t reflect the temporal reasoning capability of video reasoning segmentation models. We argue that the capability of handling complex videos, which include occluded scenes and longer duration, truly reflects the model’s temporal reasoning capacity. As shown in Fig. 2, the user input could be answered by segments that contain key information, while the rest of the video is irrelevant. Stemming from this idea, we implement Grounded-VideoLLM [52], a SOTA multi-modal LLM capable of grounding the input user queries in long videos, to serve as our Key Segment Extractor. We use a prompt specifying the user query and ask the model to output the starting and ending timestamps of the segments that describe the content of the user query. We take the average from the top-K responses to obtain the starting and ending frames of the key segments, where we denote as V_{key} , comprising T_{key} frames. Based on the key segments, we also sample T_{ref} using an adaptive global sampling strategy. We will specify the details of different sampling strategies in the appendix. The extracted frames are then fed into the encoder.

3.3. Context Synthesizer

This module aims to aggregate text-related visual features and inject them to generate text embedding representing the current frame. With the text embeddings $\mathbf{x}_{txt}$ and visual

features, context-based aggregation is formulated as:

$$\mathbf{E}_t = \text{FFN}(\text{CrossAttn}(\mathbf{x}_{t:xt}, F_t^L, F_t^L)), \quad (1)$$

$$\mathbf{E}_t^c = \text{Concat}\{\mathbf{E}_t^i\}_{i=1}^K, \quad (2)$$

where ‘CrossAttn’ refers to cross-attention operation, and $\mathbf{x}_{t:xt}$ is the query, while F_t^L is the value and key. By the cross-attention, we further synthesize the visual cues into the refined text embeddings $\mathbf{E}_t$. However, we argue that not all queries are needed for the refined text embeddings. Unlike QFormer which adopts 32 queries as input LLM tokens, we propose to condense these embeddings into more compact ones. After getting $\mathbf{E}_t$, we select the output embeddings with K highest-response scores from the attention matrix out of M . Therefore, the final input embeddings $\mathbf{E}_t^c$ preserve the most relevant visuals.

Multi-level Segmentation Tokens. To enrich the target-specific encoding and thereby facilitate the generation of high-quality mask tracklets across frames, we devise multi-scale segmentation tokens, representing both frame-level and video-level concepts, tailored to meet the characteristics of video segmentation tasks that demand the modeling of multiple target movements. Formally, we define $C_{\text{seg}} = \{c_n^s \in \mathbb{R}^d\}_{n=1}^N$, where $s \in \{f, v\}$ indicates the frame- or video-scale tokens, N denotes the number of tokens per scale, and d refers to the hidden dimension of LMMs. The multi-level segmentation tokens C_{seg} , combined with the visual features F_t^L , and condensed text embeddings $\mathbf{E}_t^c$, are then processed by the LLM to generate the response y_{out} :

$$y_{\text{out}} = \mathcal{F}(\text{Proj}_{V \rightarrow L}(F_t^L), \mathbf{E}_t^c, C_{\text{seg}}). \quad (3)$$

Using an example can provide a more vivid explanation of how Large Language Models generate responses. When the user inputs “segment the leftmost zebra in the video”, the output will include not only a textual answer but also the segmentation tokens. C_{seg} : “The zebra is c_1^f, c_1^v ” (considering the easiest case where $N = 1$ for C_{seg}). Before the decoder, an additional projection layer Φ is utilized to adjust the dimensions of segmentation tokens, in the form of $Q^s = \Phi(C_{\text{seg}})$, ($s \in \{f, v\}$). The output multi-scale segmentation embeddings Q^v, Q^f derived from C_{seg} are inputs to the pixel decoder $\mathcal{D}$ with F_t for mask tracklet generation.

3.4. Hierarchical Temporal Synchronizer

We design a novel hierarchical temporal synchronizer $\mathcal{D}$ to generate mask tracklets from the multi-scale visual features F_t and segmentation tokens Q^v, Q^f . Derived from the design of the transformer decoder in Mask2Former, $\mathcal{D}$ consists of L decoder layers, where l^{th} layer cascades a masked cross-attention h_{CA}^l , a self-attention h_{SA}^l , and a feed-forward network FFN^l . Now, we have two scales of segmentation embeddings, Q^f focusing on every frame separately, while Q^v interacting with the whole video features. Both the

frame- and video-level segmentation embeddings pass the transformer decoder, and are updated as follows:

$$\mathbf{Q}^{l+1,s} = \text{FFN}^s(h_{\text{SA}}^s(h_{\text{CA}}^s(\mathbf{Q}^{l,s}, F_t^l))), \quad (4)$$

where $s \in \{f, v\}$, and the $h_{\text{CA}}^s(q, r)$ indicates the cross-attention with query embedding q and reference embedding r . Then, we feed the frame- and video-level embeddings in aggregation in order to interchange video-level and frame-level information, and this process can be formulated as:

$$\begin{aligned} \mathbf{Q}^{l+1,v'} &= \gamma \cdot \text{Softmax}((\mathbf{Q}^{l+1,v} \times \mathbf{Q}^{l+1,f\top}) \times \mathbf{Q}^{l+1,f}) \\ &\quad + (1 - \gamma) \cdot \mathbf{Q}^{l+1,v} \end{aligned} \quad (5)$$

where $\mathbf{Q}^{l+1,v'}$ is the modulated video-level segmentation embeddings at l^{th} layer. In this aggregation process, we aggregate the frame-level embeddings with higher responses to our video-level segmentation embeddings in a momentum-based manner. The momentum factor γ , empirically set to 0.03, governs the update of the video-level embeddings. This choice is based on the assumption that the aggregation of frame-level embeddings should not induce substantial changes to the overall video-level representations.

Training Objectives. The model is trained end-to-end using text generation loss $\mathcal{L}_{\text{txt}}$ and segmentation mask loss $\mathcal{L}_{\text{mask}}$. The overall objective $\mathcal{L}$ is weighted sum of these losses, determined by λ_{txt} and λ_{mask} : $\mathcal{L} = \lambda_{\text{txt}}\mathcal{L}_{\text{txt}} + \lambda_{\text{mask}}\mathcal{L}_{\text{mask}}$, where mask loss λ_{mask} is divided into binary cross-entropy loss and dice loss for both video-level and frame-level: $\mathcal{L}_{\text{mask}}^s = \lambda_{\text{ce}}^s\mathcal{L}_{\text{ce}}^s + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}}$ ($s \in \{f, v\}$), and $\mathcal{L}_{\text{txt}}$ is auto-regressive cross-entropy loss for text generation.

3.5. VideoReasonSeg

For a more comprehensive evaluation of VRS, we establish *VideoReasonSeg*, a dataset containing text instructions, corresponding high-quality masks as well as multiple choices. To ensure reliable assessment, we collect a diverse set of videos from Youtube-VIS [69], OVIS [42], LV-VIS [55], MovieChat [49], and VideoMME [15]. Videos are categorized based on their duration, including short (< 1 minutes) and long (> 5 minutes). To fully evaluate the reasoning capabilities, we design two types of evaluation standards: 1) multiple-choice QAs; 2) instructions and answers. Considering that not all the annotations of selected datasets follow the multiple-choice QA format, we automatically convert the video annotations into this format via LLMs. This is mainly because the open-ended answers have to be scored by LLMs or user studies, which may either introduce evaluation bias or manual intervention. Ultimately, we produce 2 multiple-choice QA pairs for each of the video clips. As for the answers, we mainly focus on the quality of segmentation masks. The resulting *VideoReasonSeg* benchmark comprises a total of 3k videos with 15k object-instruction pairs. All

Table 1. **Video Reasoning Segmentation on VideoReasonSeg benchmark.** “Seg” refers to “Segmentation” while “MC” indicates “Multiple Choices”. ViLLa consistently outperforms other previous methods with different backbones and on different metrics by a large margin. Noted that previous methods are retrained to include our new proposed VideoReasonSeg in the training set.

Method	Short		Long		Overall	
	Seg $\mathcal{J}\&\mathcal{F}$	MC Accuracy	Seg $\mathcal{J}\&\mathcal{F}$	MC Accuracy	Seg $\mathcal{J}\&\mathcal{F}$	MC Accuracy
MTTR [5]	17.2	-	12.7	-	15.6	-
ReferFormer [64]	17.5	-	13.4	-	16.0	-
LMPM [12]	26.4	-	18.5	-	23.1	-
OnlineRefer [62]	26.9	-	18.7	-	16.3	-
LISA-LLaVA-7B [23]	43.8	34.5	27.3	29.5	38.4	32.1
LISA-LLaVA-13B [23]	45.0	35.2	28.1	30.7	39.2	32.9
VideoLISA*-InternVideo2-1B [67]	51.0	38.4	32.4	32.9	45.3	36.7
VideoLISA*-InternVideo2-6B [67]	54.8	43.6	33.8	34.9	47.8	39.2
VISA-LLaVA-7B [67]	48.9	37.5	30.4	31.9	43.1	35.2
VISA-LLaVA-13B [67]	54.6	40.1	33.8	33.2	47.4	37.9
VISA-Chat-UniVi-13B [67]	54.8	42.6	34.0	34.3	47.6	39.6
VISA-InternVideo2-6B [67]	57.9	44.6	36.2	35.8	50.5	40.5
ViLLa-InternVideo2-1B [59]	57.4	46.5	38.9	37.6	51.2	43.1
ViLLa-InternVideo2-6B [59]	62.6	53.8	42.0	41.0	55.4	49.9

Table 2. **Referring Video Object Segmentation on Ref-YouTube-VOS, Ref-DAVIS17, and MeViS.** ViLLa consistently outperforms other methods on different benchmarks and on different metrics by a large margin.

Method	Ref-YouTube-VOS			Ref-DAVIS17			MeViS		
	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
URVOS [47]	47.2	45.3	49.2	51.6	47.3	56.0	27.8	25.7	29.9
LBDT [13]	49.4	48.2	50.6	54.1	-	-	29.3	27.8	30.8
ReferFormer [64]	62.9	61.3	64.6	61.1	58.1	64.1	31.0	29.8	32.2
LMPM [12]	-	-	-	-	-	-	37.2	34.2	40.2
OnlineRefer [62]	62.9	61.0	64.7	62.4	59.1	65.6	-	-	-
DsHmp [18]	67.1	65.0	69.1	64.9	61.7	68.1	46.4	43.0	49.8
LISA [23]	54.4	54.0	54.8	66.0	63.2	68.8	37.9	35.8	40.0
PixelLM [45]	55.0	54.3	55.7	66.7	63.4	70.0	38.7	36.3	41.1
VideoLISA [4]	61.7	60.2	63.3	67.7	63.8	71.5	42.3	39.4	45.2
VISA [67]	63.0	61.4	64.7	70.4	67.0	73.8	44.5	41.8	47.1
ViLLa	67.5	64.6	70.4	74.3	70.6	78.0	49.4	46.5	52.3

Table 3. **Video Reasoning Segmentation on ReVOS.**

Method	ReVOS		
	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
LISA-LLaVA-13B	41.8	39.6	43.9
VISA-Chat-UniVi-13B	50.9	48.8	52.9
VISA-InternVideo2-6B	52.4	50.1	54.7
ViLLa-InternVideo2-6B	57.0	54.9	59.1

videos are partitioned into two splits: `train` and `val`, containing 2k (1820 short videos and 180 long ones) and 1k videos (880 short videos and 120 long ones) respectively. The dataset incorporates three distinct complex scenarios: multiple object segmentation (~ 800), rapid motion (~ 200), and occlusion in long videos (~ 300). The details of data annotation are given in the appendix.

Evaluation Metrics. We follow previous works [12, 47] to adopt $\mathcal{J}\&\mathcal{F}$ as the main evaluation metric, which is the average of region similarity $\mathcal{J}$ and contour accuracy $\mathcal{F}$. Besides, we adopt accuracy as the metric for multiple choices.

Dataset Generation Pipeline. We utilize GPT-4V for generating questions and multiple choices. Specifically, we feed all the instance category names and corresponding masks in the video to GPT-4V. With our carefully designed prompts, GPT-4V autonomously selects instances to construct question-answer pairs based on the video contents. Examples of such prompts are illustrated in the appendix.

4. Experiment

In this section, we first present the implementation details and the evaluation benchmark, and then show the comparison results on our proposed reasoning benchmarks and referring segmentation benchmarks. Finally, we ablate on the key components in ViLLa.

4.1. Implementation Details

We use the pre-trained multimodal model from InternVideo2-1B and 6B [59], with LoRA for fine-tuning. For the visual encoder and LLM, we apply pre-trained UMT-L from the

Table 4. **Vision backbones.** InternVideo2 demonstrates better performance (1.5 and 2.4 increase) compared to previous backbones.

Vision Backbone	Segmentation $\mathcal{J}\&\mathcal{F}$	Multiple Choices Accuracy
SAM [22]	50.5	40.5
LLaVA-7B [32]	52.6	43.1
InternViT [6]	53.9	47.5
InternVideo2 [59]	55.4	49.9

Table 6. **Hierarchical Temporal Synchronizer.** Multi-scale tokens and the corresponding synchronization provide significant improvements to the baseline method. Noted that synchronization is based on multi-scale tokens.

ID	Multi-scale Tokens	Segment Synchronization	VidReasonSeg $\mathcal{J}\&\mathcal{F}$	Accuracy	Refer-VOS $\mathcal{J}\&\mathcal{F}$
1			53.7	48.6	65.5
2	✓		54.2	48.9	65.8
3	✓	✓	55.4	49.9	67.5

InternVideo2 stage1 model as the vision encoder and Vicuna-7B v0. Following BLIP2, we deploy QFormer using the pre-trained $BERT_{base}$. To generate multi-scale visual features, we adopt the ViT-Adapter to generate the necessary multi-scale features for the hierarchical temporal synchronizer. Due to a much higher GPU memory requirement of video segmentation compared to image segmentation, an efficient version of the ViT-Adapter is applied by removing all injectors.

We use 8 A100 GPUs for training 50 epochs (approximately 1.5 days). We use the AdamW optimizer with the learning rate and weight decay set to $2e-5$ and 0.02, respectively. The batch size per device is set to 4, the input frame number is 8, the resolution is 224, and the gradient accumulation step is set to 10. Further details are presented in the appendix.

4.2. Benchmarks

Benchmarks. We evaluate ViLLa on benchmarks with video segmentation and question-answering evaluations: VideoReasonSeg, referring video segmentation (Refer-Youtube-VOS, Ref-DAVIS17), ReVOS, and video segmentation (Youtube-VIS series). For the VideoReasonSeg and referring video segmentation benchmark, we use the $\mathcal{J}\&\mathcal{F}$ and average accuracy to evaluate the video reasoning of models. In video segmentation, we formulate the queries following the dataset’s annotations in the format of “Please segment the <description>”, with <description> corresponding to the descriptions of the instance categories. Training involves random sampling from VideoReasonSeg, VQA data, Video instance segmentation, and Referring video segmentation datasets. The model follows the training protocols of PixelLM [45], pre-training on image datasets for segmentation token initialization.

Table 5. **Key component designs.** Three key modules significantly boost the baseline performance (1.7, 3.6, and 2.2) separately.

ID	Key Segment Extractor	Context Synthesizer	Hier. Temp. Synchronizer	VidReasonSeg $\mathcal{J}\&\mathcal{F}$	Accuracy	Refer-VOS $\mathcal{J}\&\mathcal{F}$
1				48.5	39.7	59.1
2	✓			50.2	43.8	60.7
3		✓		52.1	43.3	63.9
4			✓	50.7	40.9	61.5
5	✓	✓		53.7	48.6	65.5
6	✓		✓	52.2	44.8	63.0
7		✓	✓	54.0	46.0	66.2
8	✓	✓	✓	55.4	49.9	67.5

Table 7. **Hierarchical Temporal Synchronizer.** Adding aggregation layers will contribute to gains, but would diminish gradually.

Architecture	VidReasonSeg $\mathcal{J}\&\mathcal{F}$	Accuracy	Refer-VOS $\mathcal{J}\&\mathcal{F}$
Baseline	53.7	48.6	65.5
+ 2 layer frame-video aggregation	54.8	49.3	66.1
+ 3 layer frame-video aggregation	55.4	49.9	67.5
+ 4 layer frame-video aggregation	55.1	49.4	67.2

4.3. Main Results

Video Reasoning Segmentation. Tab. 1 compares ViLLa with adapted competing methods on video reasoning segmentation. For the image reasoning segmentation method, to adapt to the video segmentation, we add an extra tracker based on SAM-Track [9]. Since the original LISA only has one segmentation token, we calculate the $\mathcal{J}\&\mathcal{F}$ between the predicted tracklet and the ground truth.

As shown in Tab. 1, our method outperforms the previous image reasoning segmentation LISA by a large margin, demonstrating that our model is capable of accomplishing tasks involving complex reasoning across time. Because video reasoning, different from referring video segmentation and image reasoning segmentation, requires the model to possess *reasoning ability* based on the *whole video*. Thus, without comprehensive knowledge of the whole video, the model can not perform well. What’s more, the results also indicate that our method exceeds previous reasoning VOS methods (VideoLISA, VISA) significantly. We argue that this is because of their naive design of sampling strategy and segmentation tokens that fail to model complex trajectories. On the other hand, our model improves the association between video and frame via our specially designed context synthesis strategy and decoder structures, distinguishing our method from previous works.

Referring Video Segmentation. Tab. 2 presents the results on referring video segmentation datasets. Our ViLLa demonstrates better performances on all these datasets, surpassing VISA by 4.5, 3.9, and 4.9 in $\mathcal{J}\&\mathcal{F}$ on Ref-Youtube-VOS, Ref-DAVIS17, and MeViS dataset, respectively.

ReVOS. Tab. 3 presents the results on ReVOS. ViLLa significantly outperforms previous methods by 4.6 $\mathcal{J}\&\mathcal{F}$, which further highlights the effectiveness of our design of synthe-

USER: <VIDEO> Which sheep has a black head?



VISA: <SEG1>, <SEG2>.



ViLLa: The sheep on the right side <SEG1>.



Figure 3. **Qualitative Comparisons** between ViLLa and VISA. Compared to VISA, our ViLLa successfully segments the sheep, which demonstrates better discrimination, temporal consistency, and conversation quality.

sizer, synchronizer, and extractor, and their robustness across different reasoning benchmarks.

4.4. Ablation Studies

We conduct extensive ablation studies to reveal the contributions of each component. Unless otherwise specified, we conduct the experiments on our VideoReasonSeg dataset.

Design Choices of Backbone. We emphasize that different vision backbones are applicable in our framework. In Tab. 4, we show that InternVideo2 performs the best in both segmentation and multiple choices, potentially because it is the state-of-the-art video foundation model.

Key Components Design. Tab. 5 demonstrates the effect of our designs based on the backbone and transformer decoder. By adopting our method, we significantly improve the baseline by extracting key information, strengthening the video-frame relationship, and fusing context into text embedding. Tested on VideoReasonSeg, our context synthesizer improves the accuracy of multiple choices by 3.6 points, indicating that synthesizing context to the text embedding and keeping the textual input contextualized is beneficial to video understanding. Also, the design of synchronizer improves the segmentation performance by 1.7 $\mathcal{J}\&\mathcal{F}$, showing that applying multi-scale tokens and synchronizing their relations are helpful for better segmentation and tracking.

Hierarchical Temporal Synchronizer. Tab. 6 shows the influence of utilizing multi-scale tokens and video-frame synchronization in hierarchical temporal synchronizer. With multi-segment-scale tokens only, the baseline model achieves a 0.5 increase $\mathcal{J}\&\mathcal{F}$, while applying the aggregation brings about an increment of 1.7 $\mathcal{J}\&\mathcal{F}$. This demon-

strates that using multi-scale tokens and synchronizing them are beneficial for video segmentation tasks. In Tab. 7, we explore the effects of increasing the number of layers in the decoder. Initially adding layers will contribute to notable gains, but the gain would diminish with more decoder layers.

4.5. Qualitative Results

In Fig. 3, we provide a visual comparison with VISA. We show that ViLLa is capable of segmenting instances (sheep with a black head) in crowded scenes with heavy occlusion. On the other hand, VISA fails to track the sheep properly after occlusion with other sheep. These results demonstrate the capacity of our ViLLa to handle complicated videos.

5. Conclusion

We introduce ViLLa, an effective and efficient model that aims to reason and segment in more complex video scenarios. After analyzing the drawbacks of previous methods, we propose the key segment extractor to select pivotal frames from the video sequence, utilize the context synthesizer to generate text-related visual features, and implement the hierarchical temporal synchronizer to encourage the synchronization between video-level and frame-level segmentation tokens. Also, to help us better tune and evaluate the reasoning segmentation models, we have introduced a more challenging benchmark VideoReasonSeg, which comprises 3k video samples with both short and long videos, with both segmentation and QA as evaluations. Benefiting from our exquisite designs for complex reasoning and video modeling, ViLLa demonstrates convincing reasoning video segmentation capabilities in various video scenarios and benchmarks.

Acknowledgement. This work is supported by the National Natural Science Foundation of China (No. 62422606, 62201484), and the National Key R&D Program of China (No. 2022ZD0160101).

References

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022. 1, 2
- [2] Ali Athar, Sabarinath Mahadevan, Aljosa Osep, Laura Leal-Taixé, and Bastian Leibe. Stem-seg: Spatio-temporal embeddings for instance segmentation in videos. In *ECCV*, 2020. 2
- [3] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv:2308.12966*, 2023. 1, 2
- [4] Zechen Bai, Tong He, Haiyang Mei, Pichao Wang, Ziteng Gao, Joya Chen, Lei Liu, Zheng Zhang, and Mike Zheng Shou. One token to seg them all: Language instructed reasoning segmentation in videos. In *NeurIPS*, 2024. 1, 6
- [5] Adam Botach, Evgenii Zheltonozhskii, and Chaim Baskin. End-to-end referring video object segmentation with multi-modal transformers. In *CVPR*, 2022. 2, 6
- [6] Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, et al. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *arXiv:2404.16821*, 2024. 7
- [7] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, 2022. 2
- [8] Ho Kei Cheng and Alexander G Schwing. Xmem: Long-term video object segmentation with an atkinson-shiffrin memory model. In *ECCV*, 2022. 1
- [9] Yangming Cheng, Liulei Li, Yuanyou Xu, Xiaodi Li, Zongxin Yang, Wenguan Wang, and Yi Yang. Segment and track anything. *arXiv:2305.06558*, 2023. 7
- [10] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. Vicuna: An open-source chatbot impressing gpt-4 with 90% chatgpt quality, 2023. 1
- [11] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *arXiv:2305.06500*, 2023. 2
- [12] Henghui Ding, Chang Liu, Shuting He, Xudong Jiang, and Chen Change Loy. Mevis: A large-scale benchmark for video segmentation with motion expressions. In *ICCV*, 2023. 6
- [13] Zihan Ding, Tianrui Hui, Junshi Huang, Xiaoming Wei, Jizhong Han, and Si Liu. Language-bridged spatial-temporal interaction for referring video object segmentation. In *CVPR*, 2022. 6
- [14] Hao Fei, Shengqiong Wu, Hanwang Zhang, Tat-Seng Chua, and Shuicheng Yan. Vitron: A unified pixel-level vision llm for understanding, generating, segmenting, editing. In *NeurIPS*, 2024. 3
- [15] Chaoyou Fu, Yuhan Dai, Yondong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *CVPR*, 2025. 3, 5
- [16] Peng Gao, Jiaming Han, Renrui Zhang, Ziyi Lin, Shijie Geng, Aojun Zhou, Wei Zhang, Pan Lu, Conghui He, Xiangyu Yue, et al. Llama-adapter v2: Parameter-efficient visual instruction model. *arXiv:2304.15010*, 2023. 2
- [17] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *CVPR*, 2020. 2
- [18] Shuting He and Henghui Ding. Decoupling static and hierarchical motion perception for referring video segmentation. In *CVPR*, 2024. 6
- [19] Miran Heo, Sukjun Hwang, Jeongseok Hyun, Hanjung Kim, Seoung Wug Oh, Joon-Young Lee, and Seon Joo Kim. A generalized framework for video instance segmentation. In *CVPR*, 2023. 2
- [20] Miran Heo, Sukjun Hwang, Seoung Wug Oh, Joon-Young Lee, and Seon Joo Kim. Vita: Video instance segmentation via object token association. In *NeurIPS*, 2022. 2
- [21] Anna Khoreva, Anna Rohrbach, and Bernt Schiele. Video object segmentation with language referring expressions. In *ACCV*, 2019. 2
- [22] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *ICCV*, 2023. 3, 7
- [23] Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, Yuhui Yuan, Shu Liu, and Jiaya Jia. Lisa: Reasoning segmentation via large language model. In *CVPR*, 2024. 1, 3, 6
- [24] Bo Li, Yuanhan Zhang, Liangyu Chen, Jinghao Wang, Jingkang Yang, and Ziwei Liu. Otter: A multi-modal model with in-context instruction tuning. *arXiv:2305.03726*, 2023. 1
- [25] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *ICML*, 2023. 1, 2
- [26] Xiangtai Li, Haobo Yuan, Wei Li, Henghui Ding, Size Wu, Wenwei Zhang, Yining Li, Kai Chen, and Chen Change Loy. Omg-seg: Is one model good enough for all segmentation? In *CVPR*, 2024. 2
- [27] Xiangtai Li, Wenwei Zhang, Jiangmiao Pang, Kai Chen, Guangliang Cheng, Yunhai Tong, and Chen Change Loy. Video k-net: A simple, strong, and unified baseline for video segmentation. In *CVPR*, 2022. 2
- [28] Yanwei Li, Chengyao Wang, and Jiaya Jia. Llama-vid: An image is worth 2 tokens in large language models. In *ECCV*, 2024. 1

- [29] Huaijia Lin, Ruizheng Wu, Shu Liu, Jiangbo Lu, and Jiaya Jia. Video instance segmentation with a propose-reduce paradigm. In *ICCV*, 2021. 2
- [30] Chang Liu, Xiangtai Li, and Henghui Ding. Referring image editing: Object-level image editing via referring expressions. In *CVPR*, 2024. 1
- [31] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *CVPR*, 2024. 1
- [32] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023. 1, 2, 7
- [33] Shilong Liu, Hao Cheng, Haotian Liu, Hao Zhang, Feng Li, Tianhe Ren, Xueyan Zou, Jianwei Yang, Hang Su, Jun Zhu, et al. Llava-plus: Learning to use tools for creating multimodal agents. In *ECCV*, 2024. 3
- [34] Zhaoyang Liu, Yanan He, Wenhai Wang, Weiyun Wang, Yi Wang, Shoufa Chen, Qinglong Zhang, Zeqiang Lai, Yang Yang, Qingyun Li, Jiashuo Yu, et al. Interngpt: Solving vision-centric tasks by interacting with chatgpt beyond language. *arXiv:2305.05662*, 2023. 1, 2, 3
- [35] Jiasen Lu, Christopher Clark, Rowan Zellers, Roozbeh Motlaghi, and Aniruddha Kembhavi. Unified-io: A unified model for vision, language, and multi-modal tasks. In *ICLR*, 2022. 2
- [36] Ruipu Luo, Ziwang Zhao, Min Yang, Junwei Dong, Minghui Qiu, Pengcheng Lu, Tao Wang, and Zhongyu Wei. Valley: Video assistant with large language model enhanced ability. *arXiv:2306.07207*, 2023. 2
- [37] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Shahbaz Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv:2306.05424*, 2023. 2
- [38] OpenAI. Chatgpt: A language model for conversational ai. Technical report, OpenAI, 2023. 1, 2
- [39] OpenAI. Gpt-4 technical report, 2023. 1, 2
- [40] Baolin Peng, Chunyuan Li, Pengcheng He, Michel Galley, and Jianfeng Gao. Instruction tuning with gpt-4. *arXiv:2304.03277*, 2023. 2
- [41] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shao-han Huang, Shuming Ma, and Furu Wei. Kosmos-2: Grounding multimodal large language models to the world. *arXiv:2306.14824*, 2023. 3
- [42] Jiyang Qi, Yan Gao, Yao Hu, Xinggang Wang, Xiaoyu Liu, Xiang Bai, Serge Belongie, Alan Yuille, Philip HS Torr, and Song Bai. Occluded video instance segmentation: A benchmark. *IJCV*, 2022. 5
- [43] Lu Qi, Yi-Wen Chen, Lehan Yang, Tiancheng Shen, Xiangtai Li, Weidong Guo, Yu Xu, and Ming-Hsuan Yang. Generalizable entity grounding via assistance of large language model. *arXiv:2402.02555*, 2024. 3
- [44] Hanoona Rasheed, Muhammad Maaz, Sahal Shaji, Abdelrahman Shaker, Salman Khan, Hisham Cholakkal, Rao M Anwer, Eric Xing, Ming-Hsuan Yang, and Fahad S Khan. Glamm: Pixel grounding large multimodal model. In *CVPR*, 2024. 3
- [45] Zhongwei Ren, Zhicheng Huang, Yunchao Wei, Yao Zhao, Dongmei Fu, Jiashi Feng, and Xiaojie Jin. Pixellm: Pixel reasoning with large multimodal model. In *CVPR*, 2024. 1, 3, 6, 7
- [46] Arezoo Sadeghzadeh, Md Baharul Islam, Md Nur Uddin, and Tarkan Aydin. Arva: An augmented reality-based visual aid for mobility enhancement through real-time video stream transformation. *IEEE ACCESS*, 2024. 1
- [47] Seonguk Seo, Joon-Young Lee, and Bohyung Han. Urvos: Unified referring video object segmentation network with a large-scale benchmark. In *ECCV*, 2020. 2, 6
- [48] Pierre Sermanet, Tianli Ding, Jeffrey Zhao, Fei Xia, Debidda Dwibedi, Keerthana Gopalakrishnan, Christine Chan, Gabriel Dulac-Arnold, Sharath Maddineni, Nikhil J Joshi, et al. Robovqa: Multimodal long-horizon reasoning for robotics. In *ICRA*, 2024. 1
- [49] Enxin Song, Wenhao Chai, Guanhong Wang, Yucheng Zhang, Haoyang Zhou, Feiyang Wu, Xun Guo, Tian Ye, Yan Lu, Jenq-Neng Hwang, et al. Moviechat: From dense token to sparse memory for long video understanding. *arXiv:2307.16449*, 2023. 5
- [50] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. Stanford alpaca: An instruction-following llama model, 2023. 2
- [51] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv:2302.13971*, 2023. 2
- [52] Haibo Wang, Zhiyang Xu, Yu Cheng, Shizhe Diao, Yufan Zhou, Yixin Cao, Qifan Wang, Weifeng Ge, and Lifu Huang. Grounded-videollm: Sharpening fine-grained temporal grounding in video large language models. *arXiv:2410.03290*, 2024. 4
- [53] Hao Wang, Cheng Deng, Fan Ma, and Yi Yang. Context modulated dynamic networks for actor and action video segmentation with language queries. In *AAAI*, 2020. 2
- [54] Hao Wang, Cheng Deng, Junchi Yan, and Dacheng Tao. Asymmetric cross-guided attention network for actor and action video segmentation from natural language query. In *ICCV*, 2019. 2
- [55] Haochen Wang, Cilin Yan, Shuai Wang, Xiaolong Jiang, Xu Tang, Yao Hu, Weidi Xie, and Efstratios Gavves. Towards open-vocabulary video instance segmentation. In *ICCV*, 2023. 5
- [56] Peng Wang, An Yang, Rui Men, Junyang Lin, Shuai Bai, Zhikang Li, Jianxin Ma, Chang Zhou, Jingren Zhou, and Hongxia Yang. Ofa: Unifying architectures, tasks, and modalities through a simple sequence-to-sequence learning framework. In *ICML*, 2022. 2
- [57] Shihao Wang, Zhiding Yu, Xiaohui Jiang, Shiyi Lan, Min Shi, Nadine Chang, Jan Kautz, Ying Li, and Jose M Alvarez. Omnidrive: A holistic llm-agent framework for autonomous driving with 3d perception, reasoning and planning. *arXiv:2405.01533*, 2024. 1
- [58] Yi Wang, Yanan He, Yizhuo Li, Kunchang Li, Jiashuo Yu, Xin Ma, Xinyuan Chen, Yaohui Wang, Ping Luo, Ziwei Liu, et al. Internvid: A large-scale video-text dataset for multimodal understanding and generation. *arXiv:2307.06942*, 2023. 1
- [59] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yanan He, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang,

- et al. Internvideo2: Scaling video foundation models for multimodal video understanding. *arXiv:2403.15377*, 2024. 2, 6, 7
- [60] Yuqing Wang, Zhaoliang Xu, Xinlong Wang, Chunhua Shen, Baoshan Cheng, Hao Shen, and Huaxia Xia. End-to-end video instance segmentation with transformers. In *CVPR*, 2021. 2
- [61] Dongming Wu, Xingping Dong, Ling Shao, and Jianbing Shen. Multi-level representation learning with semantic alignment for referring video object segmentation. In *CVPR*, 2022. 2
- [62] Dongming Wu, Tiancai Wang, Yang Zhang, Xiangyu Zhang, and Jianbing Shen. Onlinerefer: A simple online baseline for referring video object segmentation. In *ICCV*, 2023. 6
- [63] Jialian Wu, Sudhir Yarram, Hui Liang, Tian Lan, Junsong Yuan, Jayan Eledath, and Gerard Medioni. Efficient video instance segmentation via tracklet query and proposal. In *CVPR*, 2022. 2
- [64] Jiannan Wu, Yi Jiang, Peize Sun, Zehuan Yuan, and Ping Luo. Language as queries for referring video object segmentation. In *CVPR*, 2022. 2, 6
- [65] Junfeng Wu, Yi Jiang, Song Bai, Wenqing Zhang, and Xiang Bai. Seqformer: Sequential transformer for video instance segmentation. In *ECCV*, 2022. 2
- [66] Junfeng Wu, Qihao Liu, Yi Jiang, Song Bai, Alan Yuille, and Xiang Bai. In defense of online models for video instance segmentation. In *ECCV*, 2022. 2
- [67] Cilin Yan, Haochen Wang, Shilin Yan, Xiaolong Jiang, Yao Hu, Guoliang Kang, Weidi Xie, and Efstratios Gavves. Visa: Reasoning video object segmentation via large language models. *arXiv:2407.11325*, 2024. 1, 3, 6
- [68] Shilin Yan, Renrui Zhang, Ziyu Guo, Wenchao Chen, Wei Zhang, Hongyang Li, Yu Qiao, Hao Dong, Zhongjiang He, and Peng Gao. Referred by multi-modality: A unified temporal transformer for video object segmentation. In *AAAI*, 2024. 2
- [69] Linjie Yang, Yuchen Fan, and Ning Xu. Video instance segmentation. In *ICCV*, 2019. 2, 5
- [70] Senqiao Yang, Tianyuan Qu, Xin Lai, Zhuotao Tian, Bohao Peng, Shu Liu, and Jiaya Jia. An improved baseline for reasoning segmentation with large language model. *arXiv:2312.17240*, 2023. 1
- [71] Qinghao Ye, Haiyang Xu, Guohai Xu, Jiabo Ye, Ming Yan, Yiyang Zhou, Junyang Wang, Anwen Hu, Pengcheng Shi, Yaya Shi, et al. mplug-owl: Modularization empowers large language models with multimodality. *arXiv:2304.14178*, 2023. 1
- [72] Kaining Ying, Qing Zhong, Weian Mao, Zhenhua Wang, Hao Chen, Lin Yuanbo Wu, Yifan Liu, Chengxiang Fan, Yunzhi Zhuge, and Chunhua Shen. Ctvis: Consistent training for online video instance segmentation. In *ICCV*, 2023. 2
- [73] Haoxuan You, Haotian Zhang, Zhe Gan, Xianzhi Du, Bowen Zhang, Zirui Wang, Liangliang Cao, Shih-Fu Chang, and Yinfei Yang. Ferret: Refer and ground anything anywhere at any granularity. *arXiv:2310.07704*, 2023. 3
- [74] Jiahui Yu, Zirui Wang, Vijay Vasudevan, Legg Yeung, Mojtaba Seyedhosseini, and Yonghui Wu. Coca: Contrastive captioners are image-text foundation models. *arXiv:2205.01917*, 2022. 2
- [75] Renrui Zhang, Jiaming Han, Aojun Zhou, Xiangfei Hu, Shilin Yan, Pan Lu, Hongsheng Li, Peng Gao, and Yu Qiao. Llama-adaptor: Efficient fine-tuning of language models with zero-init attention. *arXiv:2303.16199*, 2023. 1
- [76] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv:2205.01068*, 2022. 2
- [77] Tao Zhang, Xiangtai Li, Hao Fei, Haobo Yuan, Shengqiong Wu, Shunping Ji, Chen Change Loy, and Shuicheng Yan. Omg-llava: Bridging image-level, object-level, pixel-level reasoning and understanding. *arXiv:2406.19389*, 2024. 3
- [78] Tao Zhang, Xingye Tian, Yu Wu, Shunping Ji, Xuebo Wang, Yuan Zhang, and Pengfei Wan. Dvis: Decoupled video instance segmentation framework. *arXiv:2306.03413*, 2023. 2
- [79] Yikang Zhou, Tao Zhang, Shunping Ji, Shuicheng Yan, and Xiangtai Li. Dvis-daq: Improving video segmentation via dynamic anchor queries. In *ECCV*, 2024. 2
- [80] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv:2304.10592*, 2023. 1, 2
- [81] Xizhou Zhu, Weiye Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. In *ICLR*, 2020. 2